CHAPTER 6

BIOTECHNOLOGY AND ITS APPLICATIONS

CHAPTER ESSENCE AT A GLANCE:
Transgenic bioengineering yields insect-resistant crops, recombinant human therapeutics, RNAi pest resistance, and gene therapy.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Bt Cotton and Insecticidal Endotoxin Activation:** Soil bacterium Bacillus thuringiensis produces crystalline protein (Cry) endotoxins toxic to insects. The toxin exists as an inactive protoxin within the bacterium; upon ingestion by insect pests (bollworms, corn borers), the alkaline pH of the insect midgut solubilizes the crystals into active toxin, which binds midgut epithelial receptors creating pores that cause osmotic lysis and larval death. CryIAc and CryIIAb control cotton bollworms, while CryIAb controls corn borer.

• **RNA Interference (RNAi) Nematode Resistance:** Meloidogyne incognita infects tobacco roots causing extensive galls and crop loss. Using Agrobacterium vectors, nematode-specific genes are introduced into tobacco, transcribing both sense and antisense RNA that anneal into double-stranded RNA (dsRNA). Cellular Dicer cuts dsRNA into siRNAs, which guide the RISC complex to cleave and silence complementary nematode mRNA, preventing translation and killing the parasite.

• **Recombinant Human Insulin (Humulin) Engineering:** Human proinsulin contains three chains: Chain A (21 amino acids) and Chain B (30 amino acids) connected by disulfide bridges, plus an extra connecting C-peptide (33 amino acids) removed during maturation. In 1983, Eli Lilly synthesized two separate DNA sequences coding for A and B chains, introduced them into E. coli plasmids to produce chains independently, and joined them in vitro by disulfide bonds to yield active mature Humulin without C-peptide.

• **Clinical Gene Therapy for ADA Deficiency:** Adenosine Deaminase (ADA) deficiency causes severe combined immunodeficiency (SCID). In 1990, the first clinical gene therapy was administered to a 4-year-old girl: patient lymphocytes were isolated, grown in culture, transfected with functional ADA cDNA via a retroviral vector, and re-infused into her circulation. Periodic infusions are required because mature lymphocytes are mortal; permanent cure requires introducing functional ADA genes into bone marrow stem cells at early embryonic stages.

ESSENTIAL TAXONOMICAL FRAMEWORKS & PROCESSES

$$\text{Bt Protoxin} \xrightarrow{\text{Alkaline pH of Insect Midgut}} \text{Active Toxin} \rightarrow \text{Epithelial Pores} \rightarrow \text{Cell Lysis and Death}$$

Specific Midgut Solubilization and Toxicity of Cry Endotoxins

$$\text{Humulin Architecture} = \text{Chain A (21 aa)} \overset{\text{Disulfide Bonds}}{\longleftrightarrow} \text{Chain B (30 aa)} \quad [\text{C-peptide deleted}]$$

Recombinant Human Insulin Subunit Structure

$$\text{RNAi Silencing} = \text{dsRNA} \xrightarrow{\text{Dicer}} \text{siRNA} \xrightarrow{\text{RISC}} \text{Cleavage of target nematode mRNA}$$

Double-Stranded RNA Post-Transcriptional Gene Silencing Cascade

$$\text{ADA Gene Therapy} = \text{Patient T-lymphocytes} + \text{Functional ADA cDNA (Retrovirus)} \rightarrow \text{Immune Restoration}$$

Ex-Vivo Retroviral Transduction Protocol for ADA SCID

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Proinsulin contains an extra connecting C-peptide that is absent in mature functional insulin; Eli Lilly produced chains A and B separately without C-peptide.

• **Alert:** Bt toxin does not harm Bacillus thuringiensis because it exists as an inactive protoxin; it becomes active only in the alkaline pH of the insect midgut.

• **Alert:** RNA interference involves silencing of a specific mRNA due to a complementary double-stranded RNA (dsRNA) molecule, preventing translation.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Why does the Bt toxin protein not kill the bacterium Bacillus thuringiensis that synthesizes it, but kills insect pests? [Score: 2 Marks]

Answer & Key Valuation Steps:

• Inactive state in bacteria: Bacillus thuringiensis synthesizes the toxic protein in an inactive protoxin crystalline form, which is harmless to the bacterium. **[1 Mark]**

• Activation in insect midgut: When ingested by an insect, the alkaline pH of the midgut solubilizes the protein crystals, converting protoxin into active toxin that creates lethal epithelial pores. **[1 Mark]**

2. What was the major technological challenge overcome by Eli Lilly in 1983 during the production of recombinant human insulin (Humulin)? [Score: 2 Marks]

Answer & Key Valuation Steps:

• Insulin is naturally synthesized in humans as a pro-hormone containing an extra C-peptide that must be removed to create active mature insulin. **[1 Mark]**

• Eli Lilly synthesized two separate artificial DNA sequences for chain A and chain B, expressed them separately in E. coli, extracted the chains, and linked them in vitro by disulfide bonds to produce mature insulin directly without C-peptide. **[1 Mark]**

3. Explain the mechanism of RNA interference (RNAi) used to confer resistance against the nematode Meloidogyne incognita in transgenic tobacco plants. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Nematode-specific genes were introduced into the host tobacco plant using Agrobacterium tumefaciens vectors. **[1 Mark]**

• The DNA was introduced such that it produced both sense and antisense RNA in host plant cells; being complementary, they paired to form double-stranded RNA (dsRNA). **[1 Mark]**

• The dsRNA initiated RNAi by directing cellular Dicer and RISC complexes to silence the specific mRNA of the nematode, preventing protein synthesis and killing the parasite when it feeds on tobacco roots. **[1 Mark]**

4. Describe the gene therapy protocol administered to treat Adenosine Deaminase (ADA) deficiency. Why is this treatment not permanent unless performed at early embryonic stages? [Score: 3 Marks]

Answer & Key Valuation Steps:

• Lymphocytes from the blood of an ADA-deficient patient are extracted and grown in laboratory cell culture. **[1 Mark]**

• A functional, normal human ADA cDNA is introduced into these cultured lymphocytes using a disarmed retroviral vector, and the engineered cells are re-infused into the patient. **[1 Mark]**

• Because mature lymphocytes are mortal cells with a finite lifespan, the patient requires periodic infusions; if the normal gene is introduced into bone marrow stem cells at early embryonic stages, it provides a permanent cure. **[1 Mark]**

5. What are transgenic animals? Explain two major applications of transgenic animals in medical research and human nutrition, citing transgenic cow Rosie. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Transgenic animals: Animals whose genome has been genetically altered by the introduction of foreign recombinant genes. **[1 Mark]**

• Nutritional enhancement (Rosie cow): In 1997, the first transgenic cow Rosie produced human protein-enriched milk containing 2.4 grams of human alpha-lactalbumin per liter, providing a more balanced natural nutrition for human babies. **[1 Mark]**

• Vaccine safety testing: Transgenic mice are developed for testing the safety of new human vaccines (e.g., polio vaccine) before human clinical trials. **[1 Mark]**

6. Name the specific Bt toxin genes engineered to control: (a) Cotton bollworms, and (b) Corn borer, and explain how the toxin causes death of the insect larva. [Score: 3 Marks]

Answer & Key Valuation Steps:

• (a) Cotton bollworm control: Coded by genes cryIAc and cryIIAb. **[0.75 Mark]**

• (b) Corn borer control: Coded by gene cryIAb. **[0.75 Mark]**

• Mechanism of insect death: Activated toxin binds to surface receptors of midgut epithelial cells, creating open pores that cause cell swelling and lysis, leading to gut perforation and death of the insect larva. **[1.5 Marks]**

7. A child suffers from severe combined immunodeficiency due to Adenosine Deaminase (ADA) deficiency. Contrast the therapeutic efficacy of enzyme replacement therapy (periodic ADA injection) with retroviral-mediated ex vivo gene therapy and in vivo embryonic gene therapy. [Score: 4 Marks]

Answer & Key Valuation Steps:

• Enzyme replacement therapy: Involves periodic intramuscular injections of purified functional ADA enzyme; it treats symptoms temporarily but requires lifelong repeated administration and is expensive without curing the defect. **[1.5 Marks]**

• Retroviral-mediated ex vivo gene therapy: Introduces functional ADA cDNA into patient lymphocytes using retroviral vectors; significantly restores cellular immunity, but because mature circulating T-cells have a limited lifespan, the patient requires regular infusions of genetically corrected cells. **[1.5 Marks]**

• Embryonic stem cell gene therapy: Involves introducing the functional ADA gene into bone marrow stem cells during early embryonic development; this delivers a permanent, lifelong cure because all subsequent hematopoietic lymphoid lineages inherit the normal active gene. **[1 Mark]**